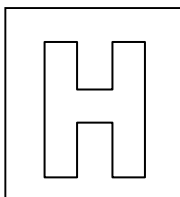


Candidate Name: _____

Class Adm No

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2018 Preliminary Exams Pre-University 3

BIOLOGY

9744/01

Paper 1 Multiple Choice

19 September 2018

1 hour

Additional Materials: Optical Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, Adm No. and class on all the papers you hand in.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Each correct answer will score one mark. A mark will not be deducted for wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

1. The first description of cell was attributed to microscopist, Robert Hooke, who was observing a slice of cork under the microscope. He termed his observation, *cella*, a latin word meaning 'a small room'. This observation later helped establish the cell theory.

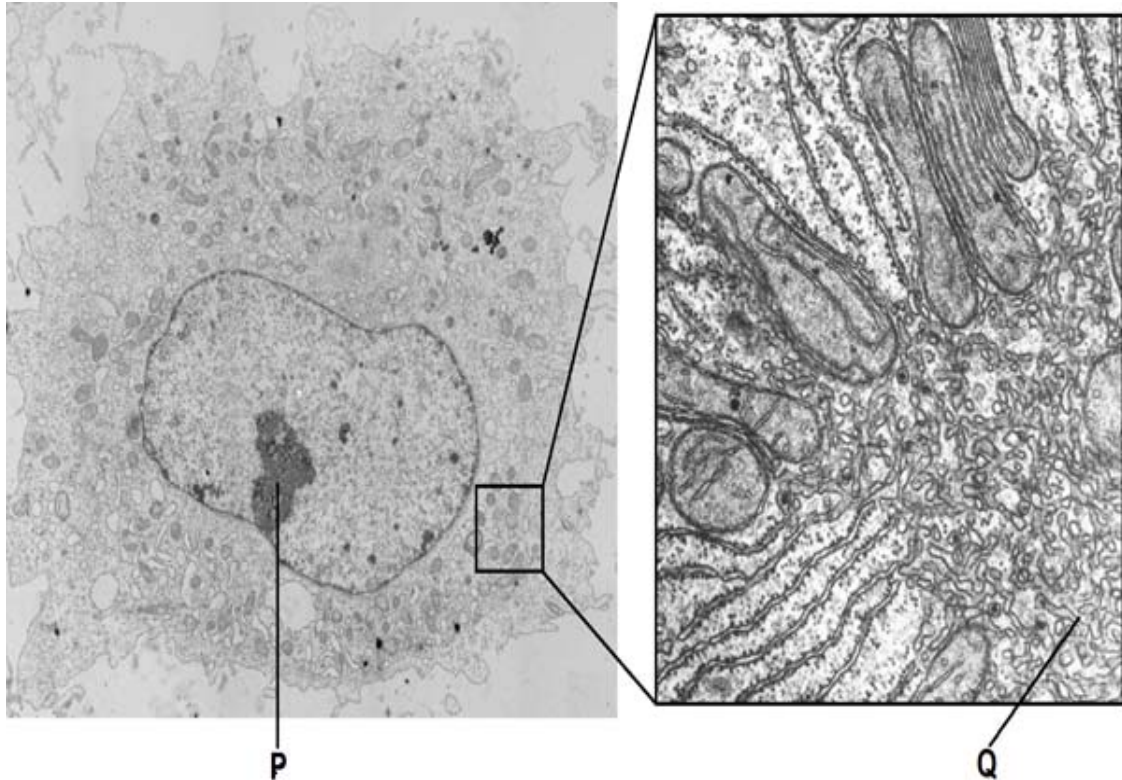
Which of the following is correct about the cell theory?

- A The theory suggests the cells are produced spontaneously.
 - B The theory agrees that virus follows the definition of cell.
 - C The theory was developed based on Robert Hooke's observation only.
 - D The theory could not be applied to all types of tissues.
-
2. The endosymbiotic theory states that mitochondria and chloroplasts originate from primitive bacterial cell. The symbiotic relationship is established when two different species benefit from each other. Further study suggests that mitochondria and chloroplast originate from aerobic and photosynthetic prokaryote respectively.

Which of the following biomolecule found in mitochondria and chloroplast best suggests that they once originate from prokaryote?

- A Carbohydrate
- B Lipid
- C Protein
- D Nucleic acid

3. The following diagram on the left shows the electro-micrograph of a cell found in a healthy individual. A region of the cell, indicated by the box, is magnified and shown in the diagram on the right.



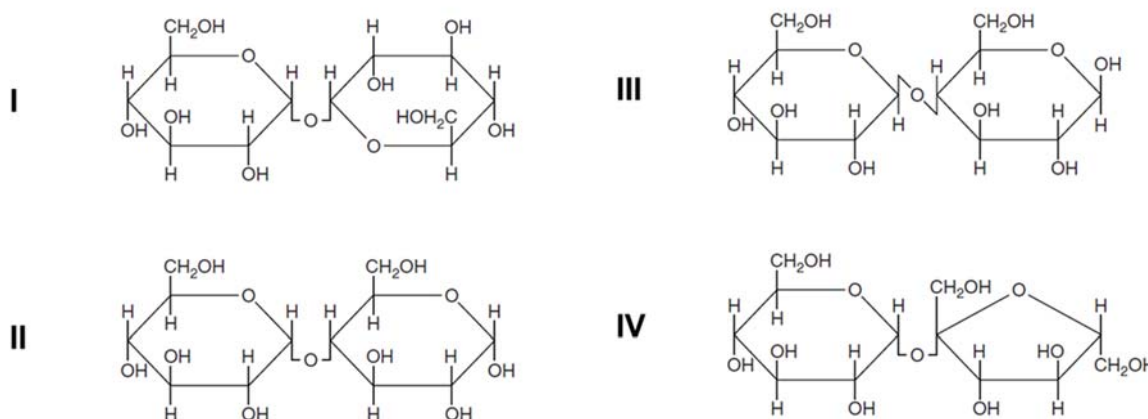
Which of the following shows the correct identities and functions of **P** and **Q**?

	Identity of P	Function of P	Identity of Q	Function of Q
A	Nucleosome	Transcription of rRNA	Smooth ER	Detoxification
B	Nucleosome	Assembly of ribosomes	Secretory vesicles	Storage of Ca^{2+}
C	Nucleolus	Assembly of ribosomes	Smooth ER	Synthesis of lipids
D	Nucleolus	Transcription of rRNA	Secretory vesicles	Modification of protein

4. Which row shows features of a typical prokaryotic cell?

	Cell size	Nucleus	DNA	Ribosome size
A	>10 μm	Absent	Circular	70S
B	>10 μm	Present	Linear	80S
C	<10 μm	Absent	Circular	70S
D	<10 μm	present	linear	80S

5. Polysaccharide, disaccharide and monosaccharides are different forms of carbohydrates. Cellobiose, maltose, sucrose and trehalose are four different disaccharides. The following diagram shows the molecular structure of these disaccharides.

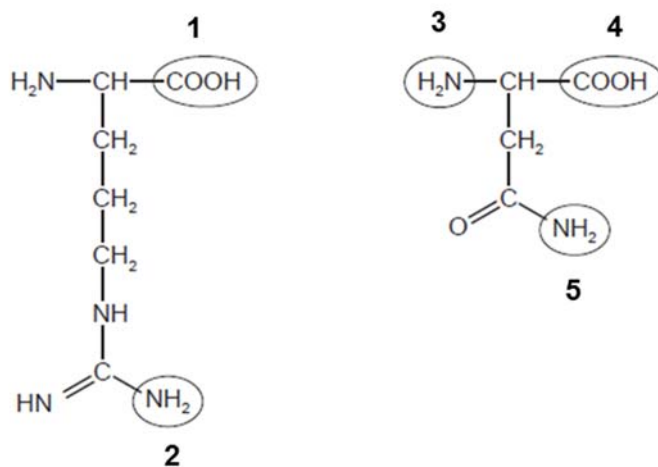


Using the information below, which option correctly identifies disaccharides I to IV?

- Cellobiose is formed from the hydrolysis of cellulose.
- Hydrolysis of cellobiose produces β -glucose.
- Trehalose is synthesised from two α -glucose.
- Maltose is formed from the hydrolysis of amylose.

	Cellobiose	Maltose	Sucrose	Trehalose
A	III	I	IV	II
B	III	II	IV	I
C	IV	I	III	II
D	IV	II	III	I

6. The diagram shows the structures of two amino acids, each of which has more than one amine (-NH_2) group.

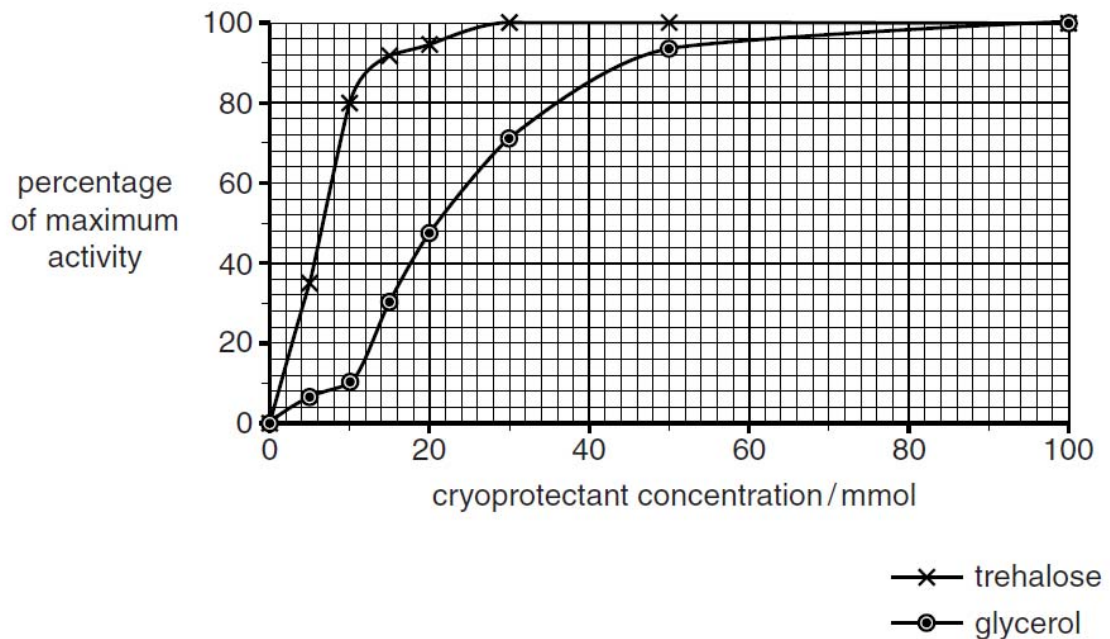


Which of the two groups form the peptide bond that holds the two amino acids?

- A 1 and 3
- B 1 and 5
- C 2 and 4
- D 2 and 5

7. An investigation was carried out to find out the protective effect of different concentrations of two cryoprotectants, trehalose and glycerol, on the activity of an enzyme. The enzyme was first incubated at freezing temperature, followed by incubating at its optimum temperature. The percentage of maximum activity of the enzyme was measured at its optimum temperature.

The following diagram shows the results of the investigation.

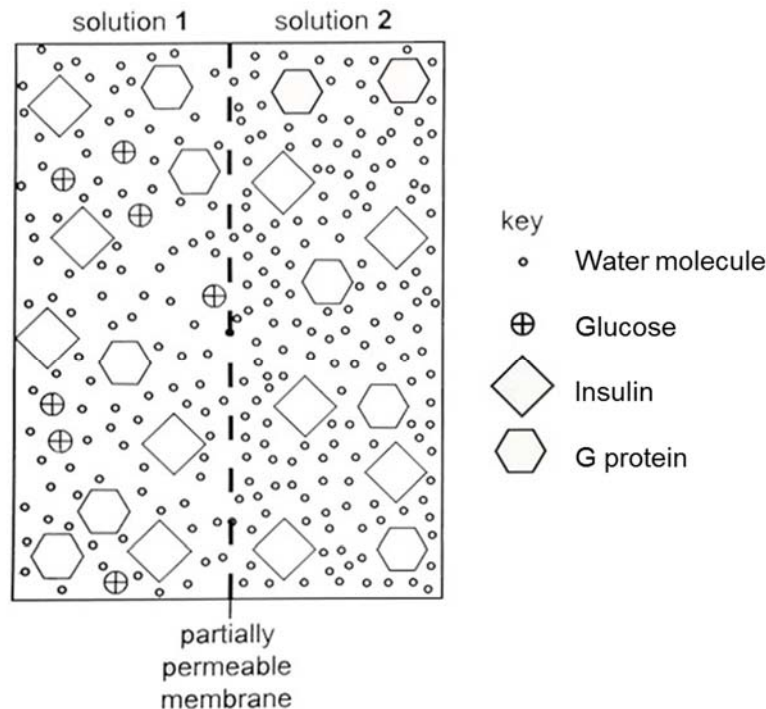


Which of the statement correctly describes the investigation?

- A The enzymes were denatured after incubating at freezing temperature.
- B Glycerol is a more effective cryoprotectant than trehalose.
- C In the absence of cryoprotectant, the enzyme remains inactive.
- D At all concentrations, trehalose results in a higher percentage of maximum enzyme activity.

8. The diagram represents an artificial partially permeable membrane separating two solutions, **1** and **2**.

The relative diameters and concentrations of the ions and molecules in the solutions and the pore diameter of the artificial partially permeable membrane are drawn proportionally in the diagram.



Which statement about the movement of the molecules between solution **1** and **2** is correct?

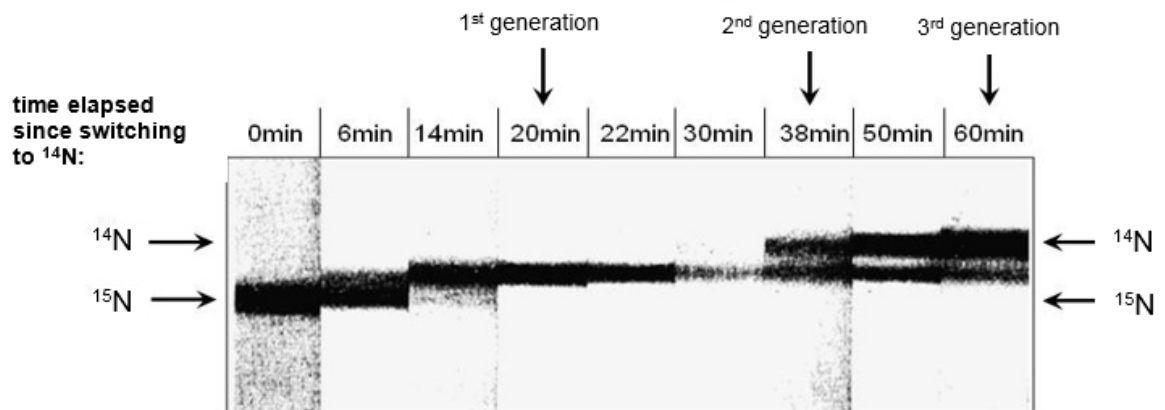
- A** The difference in water potential causes water to move by osmosis from solution **1** to solution **2**, a diffusion gradient causes net movement of glucose from solution **1** to solution **2**, and insulin and G protein to not move between the two solutions.
- B** Solution **1** has a lower water potential than solution **2**, causing net movement of water from solution **2** to solution **1** until the water potentials are the same; all the other molecules are too large to pass through the artificial partially permeable membrane.
- C** Insulin and G protein pass through the artificial partially permeable membrane equally in both directions because they are of the same concentration in solution **1** and solution **2**, but glucose and water diffuse in opposite directions until the concentrations of solution **1** and solution **2** are equal.
- D** Net movement of water occurs down a water potential gradient from solution **2** to solution **1**, net movement of glucose occurs down a diffusion gradient from solution **1** to solution **2**, and no movement of insulin and G protein occurs across the artificial partially permeable membrane.

9. In an investigation to study the mode of DNA replication, *Escherichia coli* (*E. coli*) cells were grown in a nutrient medium containing heavy isotope of nitrogen (^{15}N) for an extended period of time until all the DNA was labelled.

These *E. coli* cells were then transferred to a nutrient medium containing only light isotope of nitrogen (^{14}N) and were allowed to multiply over three generations. The DNA of the *E. coli* cells was then harvested at nine different time intervals.

Subsequently, density gradient centrifugation of these *E. coli* DNA using caesium chloride was performed.

The diagram shows the results obtained.



Which statements are consistent with the results observed?

- I There was no evidence of semi-conservative DNA replication.
 - II In the 1st generation, only hybrid $^{14}\text{N}/^{15}\text{N}$ DNA was produced.
 - III In the second generation, 50% of $^{14}\text{N}/^{15}\text{N}$ DNA and 50% light $^{14}\text{N}/^{14}\text{N}$ DNA were produced.
 - IV In the 3rd generation, 75% hybrid $^{14}\text{N}/^{15}\text{N}$ DNA and 25% light $^{14}\text{N}/^{14}\text{N}$ DNA were produced.
 - V In the subsequent 4th generation, it is impossible to deduce the amount of DNA with ^{15}N .
- A I and II only
 - B II and III only
 - C III and IV only
 - D II, III and IV only

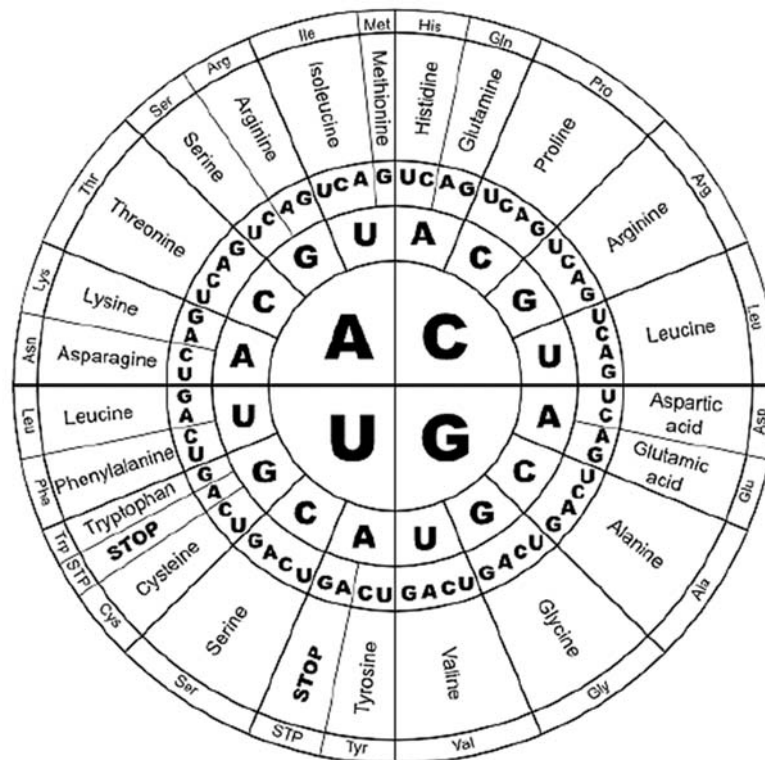
10. What is the minimum number of hydrogen bonds in a length of DNA containing 800 nucleotides?

A 800
B 1200
C 1600
D 2400

11. The first five DNA triplets that code for a particular protein is shown below:

3' CAC GGA AGC CCA GAA 5'

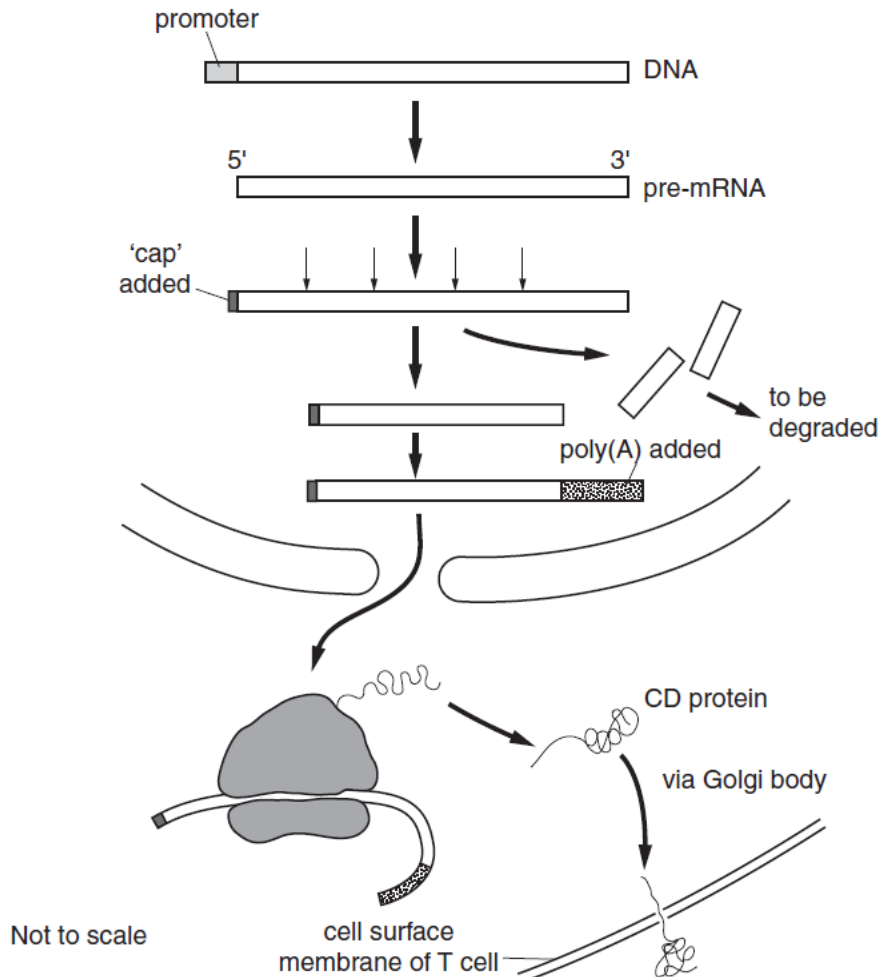
The genetic information in the sequence above is eventually converted into a specific amino acid sequence according to the figure below.



Using the given information, what is the sequence of the protein encoded by the DNA sequence above?

A His – Gly – Ser – Pro – Glu
B Val – Pro – Ser – Gly – Leu
C Lys – Thr – Arg – Arg – His
D Phe – Try – Ala – Ser – Val

12. The name of T lymphocytes is coined from the word 'thymus', the gland where the T lymphocyte matures. In the thymus, the T lymphocyte undergoes differentiation and maturation to form functional T lymphocytes. During differentiation, specific cell surface protein known as CD proteins are produced and inserted into the cell surface membrane.



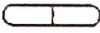
Which of the statement is wrong about the production of CD protein?

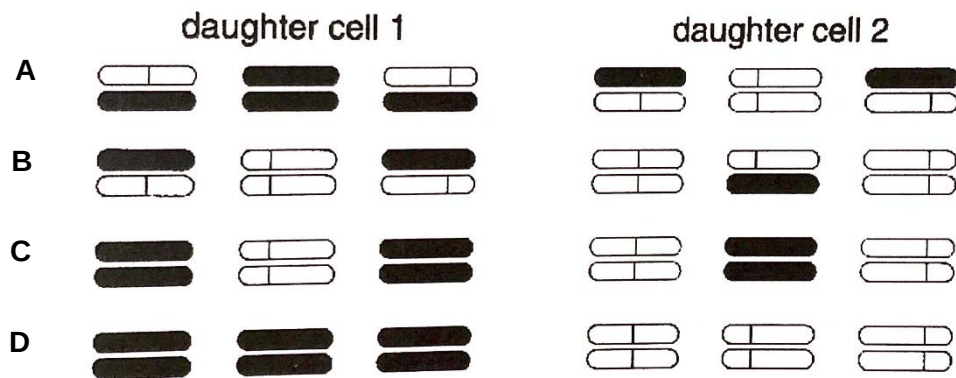
- A The ribosome synthesising the CD protein is attached to the rough endoplasmic reticulum.
- B The promoter on the DNA is the start site of protein synthesis.
- C CD protein must be processed in the Golgi body before insertion in the cell surface membrane.
- D The type of CD protein inserted into the cell surface membrane determines the fate of the T lymphocyte.

13. The following experiment was carried out.

- 1 Haploid cells, containing three chromosomes each, were grown in a medium containing radioactive thymine, so that all the DNA was labelled.
- 2 Cells in early interphase were then transferred to a medium where the available thymine was not radioactive.
- 3 A single cell was immediately isolated and allowed to divide once. When the two daughter cells reached the next metaphase, they were fixed and their three chromosomes were inspected for radioactivity.

Which diagram represents the distribution of radioactivity at metaphase in the two daughter cells?

key  = normal chromatid  = radioactive chromatid



14. Which of the following statement(s) are true about the operon systems in prokaryote?

- I Inducible operons can be switched on in the presence of regulatory protein that acts as repressor.
- II Repressible operons are usually actively undergoing gene expression but are switched off in the presence of regulatory protein that acts as repressor.
- III The co-repressor in trp operon is the trp repressor that binds to the operator of the trp operon.
- IV The binding of trp repressors to the operators is reversible and the trp repressor is an allosteric protein.
- V As the level of tryptophan increases, the repressors are able to bind to the operator to inhibit transcription of trp genes.

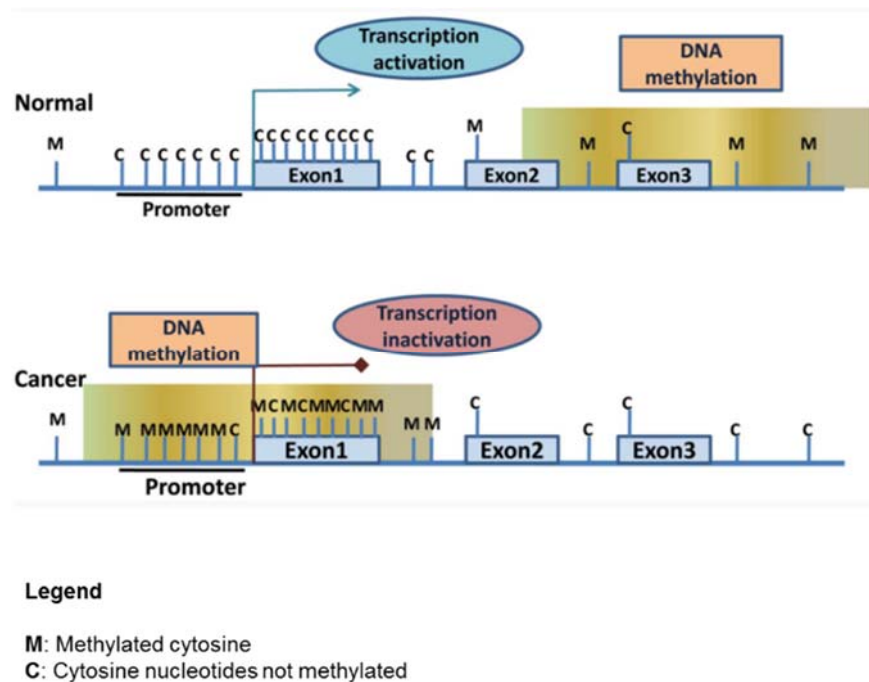
- A I, III and IV only
- B II, III and V only
- C I, IV and V only
- D II, IV and V only

15. Which option correctly compares the differences between gene expression in prokaryote and eukaryote?

	Transcriptional Control		Translational Control	
	Prokaryote	Eukaryote	Prokaryote	Eukaryote
A	RNA polymerase can only bind to promoter in the presence of transcription factors.	RNA polymerase can bind directly to promoter even without transcription factors.	Polycistronic mRNA directs the synthesis of several polypeptides.	Monocistronic mRNA directs synthesis of one polypeptide.
B	RNA polymerase can bind directly to promoter even without transcription factors.	RNA polymerase can only bind to promoter in the presence of transcription factors.	Polycistronic mRNA directs the synthesis of several polypeptides.	Monocistronic mRNA directs synthesis of one polypeptide.
C	RNA polymerase can only bind to promoter in the presence of transcription factors.	RNA polymerase can bind directly to promoter even without transcription factors.	Monocistronic mRNA directs synthesis of one polypeptide.	Polycistronic mRNA directs the synthesis of several polypeptides.
D	RNA polymerase can bind directly to promoter even without transcription factors.	RNA polymerase can only bind to promoter in the presence of transcription factors.	Monocistronic mRNA directs synthesis of one polypeptide.	Polycistronic mRNA directs the synthesis of several polypeptides.

16. DNA methylation is a common modification in mammals, plants and even bacteria. It occurs during DNA replication and involves covalent bond formation. In mammals, DNA methylation occurs mainly in CpG islands, regions of nucleotides with high frequency of cytosine and guanine nucleotides. DNA methylation may affect cell differentiation and cell cycle progression. This implies that DNA methylation patterns are often tissue-specific, affecting certain cells under different situations.

The following diagram shows the difference in DNA methylation pattern in normal and cancer cells.



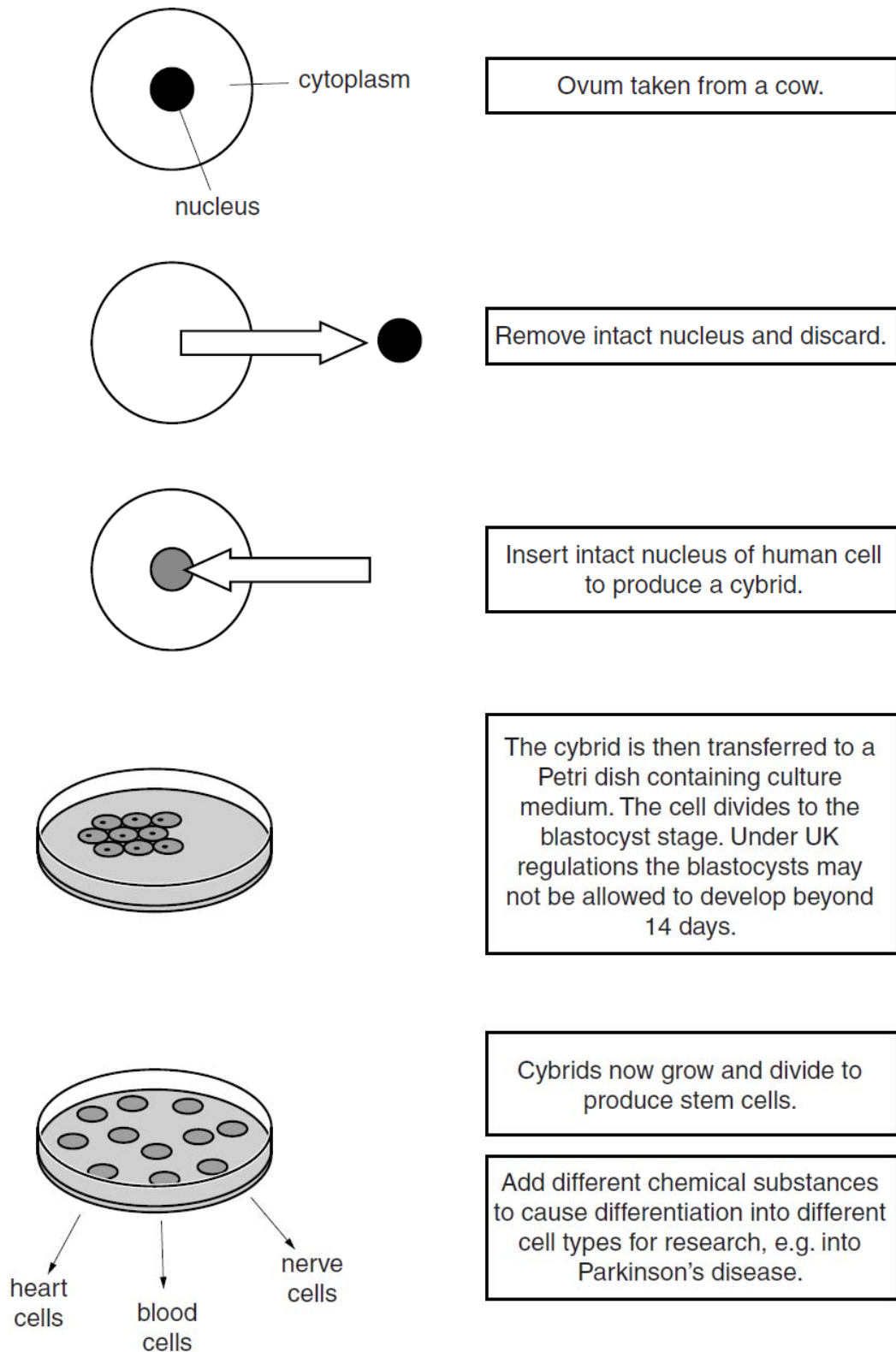
Chen, Q.W., Zhu, X.Y., Li, Y.Y., & Meng, Z.Q. (2014). Epigenetic regulation and cancer (Review). *Oncology Reports*, 31, 523-532. <https://doi.org/10.3892/or.2013.2913>

Which of the following statement is true about DNA methylation in normal and cancer cells?

- A In normal cells, CpG islands in active promoters are not methylated and thus lead to transcriptional activation. However, in cancer cells, CpG islands in promoter are methylated and thus inhibit transcriptional activation.
- B In normal cells, CpG islands in active promoters are methylated and thus lead to transcriptional activation. However, in cancer cells, CpG islands in promoter are not methylated and thus inhibit transcriptional activation.
- C In cancer cells, CpG islands in active promoters are methylated and this changes the heterochromatic structure to euchromatic structure.
- D In cancer cells, CpG islands in active promoters are methylated and this allows the expression of downstream genes coding for cell proliferation.

Use the following information to answer question **17** and **18**.

A cybrid is a cytoplasmic hybrid cell. The process of producing the cybrid cells is shown in the following diagram.



17. Which of the following statements about the cybrid cell are true?

- I** Genes expressed in the blastocyst are of cow and human origin.
- II** The blastocysts formed from the cybrid cell contains genome from the cow.
- III** Chemical substances such as transcription factors are used to induce differentiation of stem cells.
- IV** The cybrid cells will not be rejected during implantation into human host.

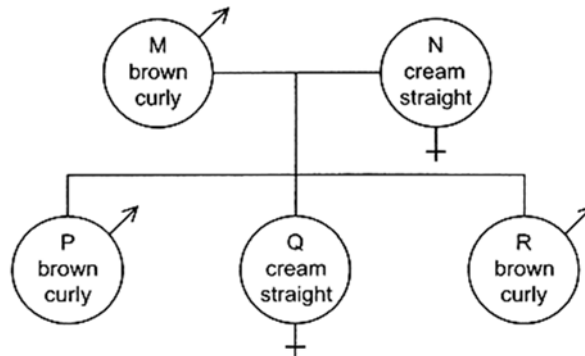
- A** I and II only
- B** I and III only
- C** II and III only
- D** III and IV only

18. Which of the following statement does not agree that the production of cybrid is an ethical procedure?

- A** The restriction of growing the blastocyst beyond 14 days because the brain and spinal cord are formed after 14 days.
- B** A human life is not destroyed during the production of cybrid cell.
- C** The presence of both human and animal genome implies that the resulting cell is not natural.
- D** Able to produce a higher number of stem cells as compared to the extraction of embryonic stem cells.

19. Assume that in goats a pair of alleles is responsible for the inheritance of hair colour and that another pair controls hair texture. These pairs are located on different autosomes. The allele for brown hair (B) is dominant to the allele for cream hair (b), and the allele for curly hair (C) is dominant to the allele for straight hair (c).

The diagram shows a cross between two goats.



If **P** is mated with a female goat of the same genotype as **M**, what are the chances of the first offspring being a male with cream coloured, straight hair?

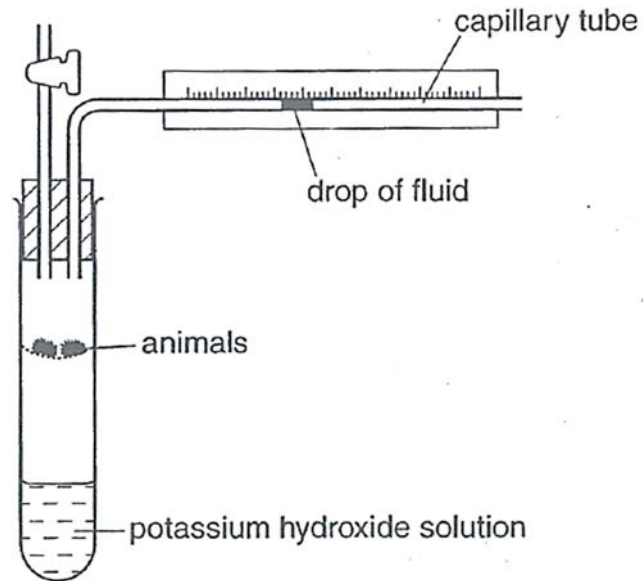
- A 0
 B 1 in 4
 C 1 in 16
 D 1 in 32
20. Possession of white or coloured feathers in poultry is controlled by two genes, A/a and B/b. Two birds that are heterozygous at both loci were crossed and the results are shown in the following table.

gametes	AB	Ab	aB	ab
AB	white feathers	white feathers	white feathers	white feathers
Ab	white feathers	white feathers	white feathers	white feathers
aB	white feathers	white feathers	coloured feathers	coloured feathers
ab	white feathers	white feathers	coloured feathers	white feathers

What best explains the inheritance of white and coloured feathers?

- A recessive epistasis in which the epistatic allele is **A**
 B recessive epistasis in which the epistatic allele is **B**
 C dominant epistasis in which the epistatic allele is **A**
 D dominant epistasis in which the epistatic allele is **B**

21. A respirometer was set up to investigate the rate of respiration of animals as shown below.

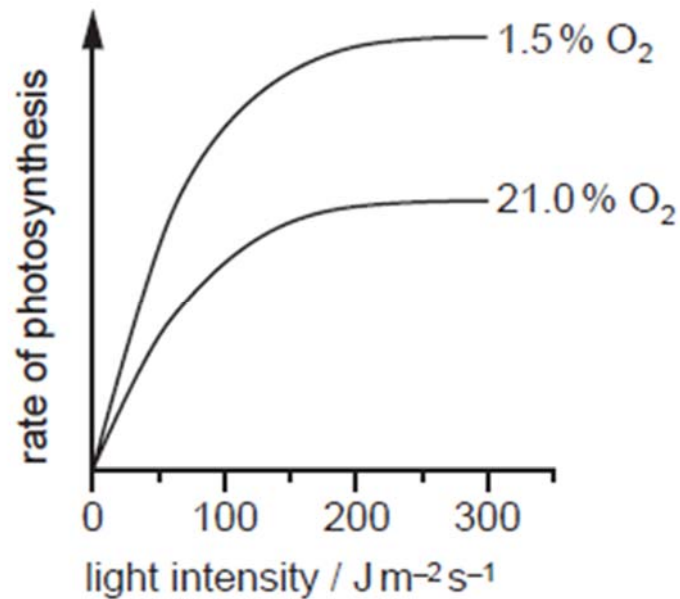


The results show that the drop of fluid moved to the left significantly more than expected. Mitochondria from the animals were isolated to investigate the cause.

Which of the following could best explain the results?

- A ATP synthase is hyperactive.
- B Higher than normal amount of grana observed.
- C Proton pump is decoupled with the electron carriers.
- D Cytochrome *b* in the electron transport chain was non-functional.

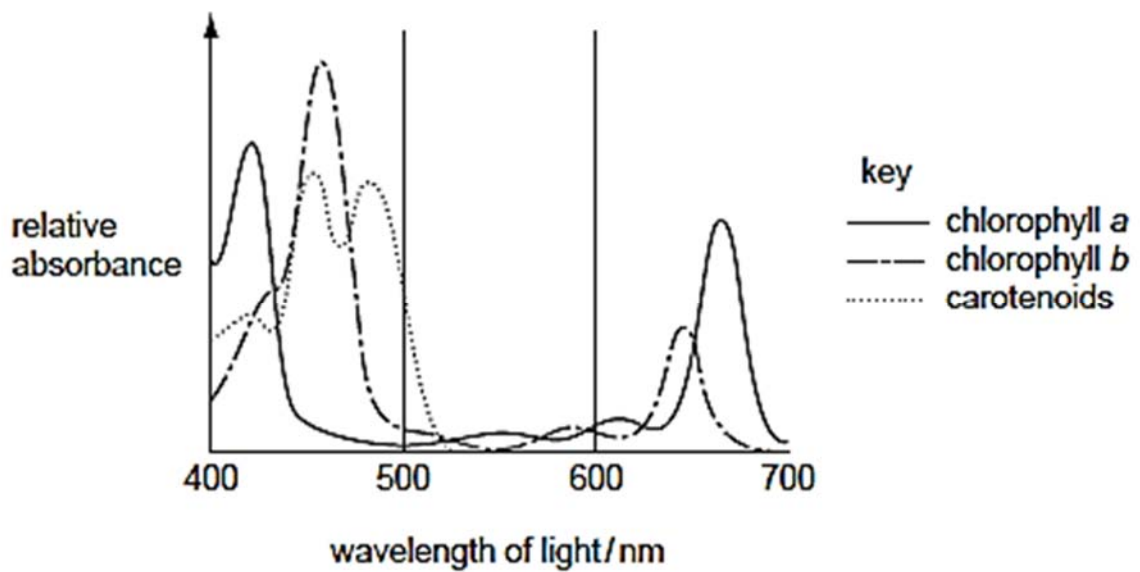
22. The following graph shows the effect of light intensity on the rate of photosynthesis of a plant at two different concentrations of oxygen. The investigation was made at atmospheric carbon dioxide concentrations.



Which of the following statement best describes the investigation?

- A The higher the light intensity, the higher the rate of photosynthesis.
- B Level of oxygen produced is the only way to measure the rate of photosynthesis.
- C The investigation should be performed at the optimal temperature to produce valid results.
- D The concentration of oxygen must be 1.5% to produce maximum rate of photosynthesis.

23. The following graph shows the absorption spectra of photosynthetic pigments.



Which statements are correct?

- I Having several pigments increase the efficiency of photosynthesis.
- II Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
- III Photosynthesis will be fastest in light at the red end of the spectrum as red light has higher energy than blue light.
- IV Chlorophyll is broken down during leaf fall, leaving the leaves to appear yellow or red due to carotenoid.

- A I and II only
- B I and III only
- C I, II and IV only
- D II and IV only

- 24.** Cystic fibrosis (CF) is a fatal genetic disease that affects the lung and causes a variety of symptoms such as persistent cough, bowel disturbance and infertility. There is currently no cure for the disease.

Approximately 1 in 25 people of Northern European descent carry the most common allele causing CF. The mutant allele expressed a protein with one missing amino acid as compared to the normal allele. To have CF, a child must inherit 1 copy of the CF gene mutation from each parent.

Gut cells with a single copy of this mutant allele took up 80% fewer *S. typhi* than normal cells. The bacterium causes the gut infection typhoid fever.

Which suggestion could provide an explanation for the number of people in Northern Europe carrying this mutant allele?

- A** The selective advantage of the heterozygotes prevents the selective pressure against individuals with CF from removing the recessive allele from the population.
- B** The selective disadvantage of the heterozygotes prevents the selective pressure against individuals with CF from removing the recessive allele from the population.
- C** The selective pressure against individuals with CF and the selective advantage of the heterozygotes are removing the dominant allele from the population.
- D** The selective pressure against individuals with CF and the selective disadvantage of the heterozygotes are removing the recessive allele from the population.

25. Speciation takes the form of either allopatric speciation or sympatric speciation.

Which sequence of events is considered most likely to lead to speciation?

A	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	allopatric speciation
B	adaptation of population	→	competition and predation leading to natural selection	→	behavioural isolation	→	sympatric speciation
C	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	sympatric speciation
D	competition and predation leading to natural selection	→	geographical isolation	→	adaptation of isolated populations	→	allopatric speciation

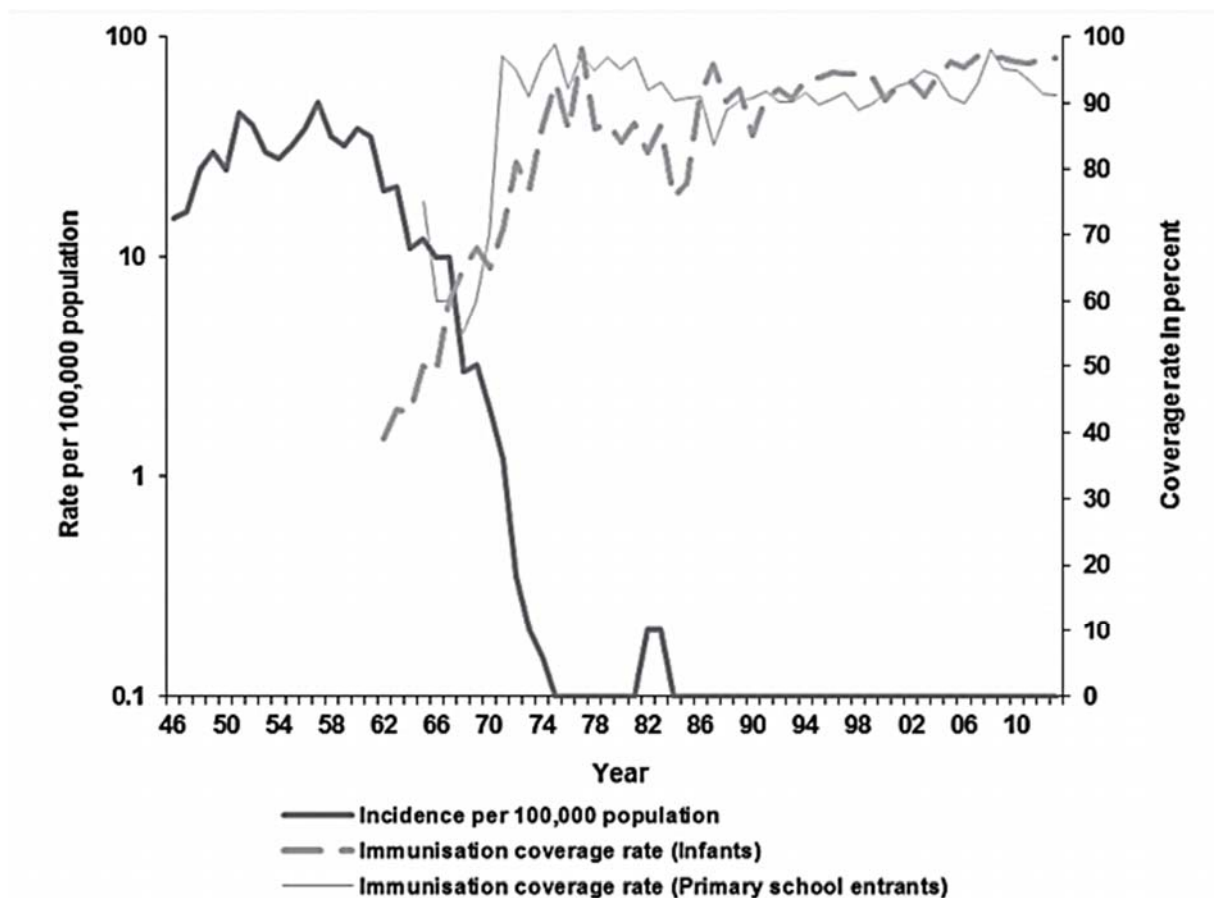
26. Which of the following event occurs after glucagon binds to the specific receptor?

- A** Glucagon binds to the extracellular binding site of G-protein coupled receptor and leads to auto-phosphorylation of the cytoplasmic tail of the receptor.
- B** Glucagon binds to the extracellular binding site of receptor tyrosine kinase, recruiting the G protein and activates adenylate cyclase.
- C** Glucagon binds to the extracellular binding site of G-protein coupled receptor and changes the conformation of G-protein coupled receptor, activating the receptor.
- D** Glucagon binds to the extracellular binding site of receptor tyrosine kinase, causing the two kinases to form a dimer.

Use the following information to answer question 27 and 28.

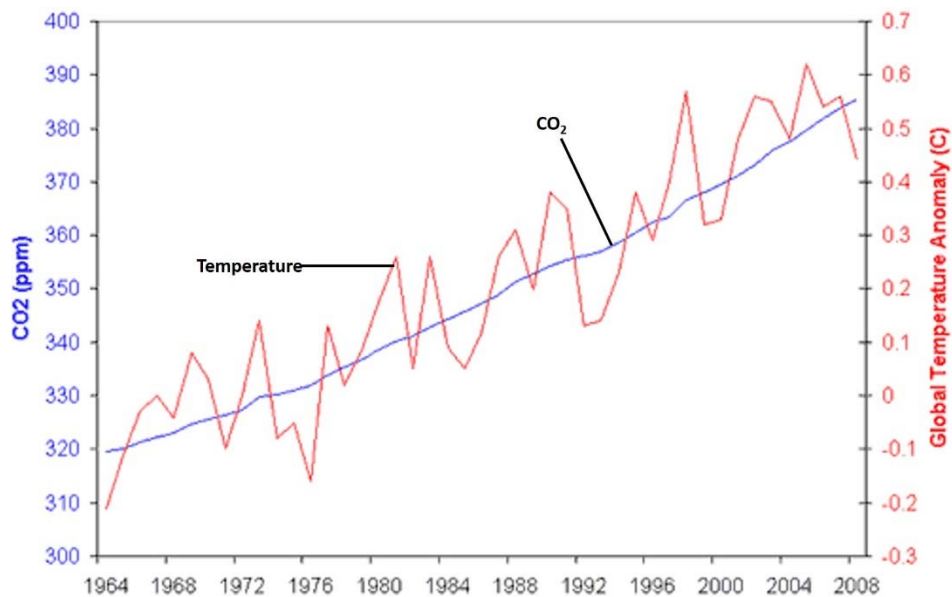
The following graph shows the incidence per 100 000 populations from diphtheria and immunisation coverage rate in Singapore from 1946 to 2013.

In Singapore, immunisation against diphtheria is compulsory by law. A child received the first dose at the age of 3 months, second dose at the age of 4 months and the third dose at the age of 5 months. The first booster dose is then given at the age of 18 months, followed by a second booster at the age of 11 years old.



- 27.** Which statement best describes the changes in incidence rate and immunisation rate?
- A** The rate of diphtheria infection decreases as the immunisation coverage rates increase.
 - B** The immunisation coverage rate in infants directly correlates with the immunisation coverage rate in primary school entrants.
 - C** The incidence of diphtheria infection follows a seasonal trend.
 - D** The immunisation coverage rate can never achieve 100%.
- 28.** Which statement best describes the immunisation against diphtheria?
- A** Immunisation is a form of passive immunity.
 - B** Herd immunity was established and this made the disease impossible to spread.
 - C** Individual can still be protected from diphtheria without the booster shot.
 - D** Booster shot is no longer necessary after 11 years ago as the antibody produced can last a lifetime after 11 years old.

29. The following graph shows the global CO₂ levels and global temperature anomaly from 1964 to 2008. Global temperature anomaly compares that year's average temperature to a long term average.



The following are some statements made by some students on the global CO₂ levels and global temperature anomaly.

Student **A**: Between 1964 to 2008, the highest temperature recorded must be in 2006.

Student **B**: Between 1964 to 2008, the global temperature fluctuates from -0.2°C to 0.45°C.

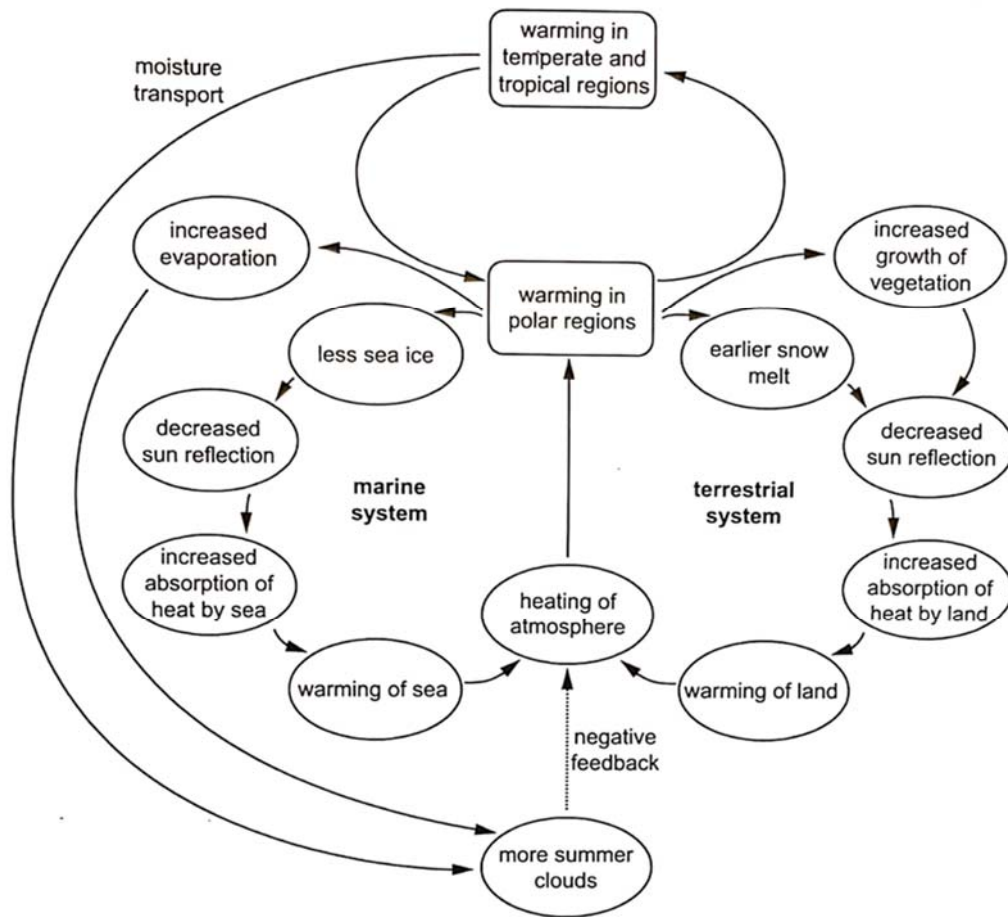
Student **C**: Between 1964 to 2008, the global CO₂ level increases from 320 ppm to 385 ppm.

Student **D**: Between 1964 to 2008, increase in temperature leads to the increase in global CO₂ level.

Which students made the correct deductions?

- A** Student **A** and **B**
- B** Student **B** and **C**
- C** Student **C** and **D**
- D** Student **B** and **D**

30. The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures in the polar regions could increase global warming?

- A Increased evaporation leads to more rainfall, which absorbs heat from the land and sea.
- B Melting of sea ice causes more cloud formation which increases absorption of heat in the atmosphere.
- C Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.
- D Melting of ice and snow results in less reflection of sunlight and more heat absorption by the earth.

End of Paper